An algorithmic approach to reducing unexplained pain disparities in underserved populations



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简介



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据已发表论文归纳，其研究聚焦群体差异的统计测量及健康与公共卫生建模，关注数据和测量方式怎样影响判断。

2020 年以第一作者发表《交通执法中的群体差异分析》（Nature Human Behaviour）；2021 年以共同第一作者发表《人群流动与传染病传播建模》（Nature）。

背景

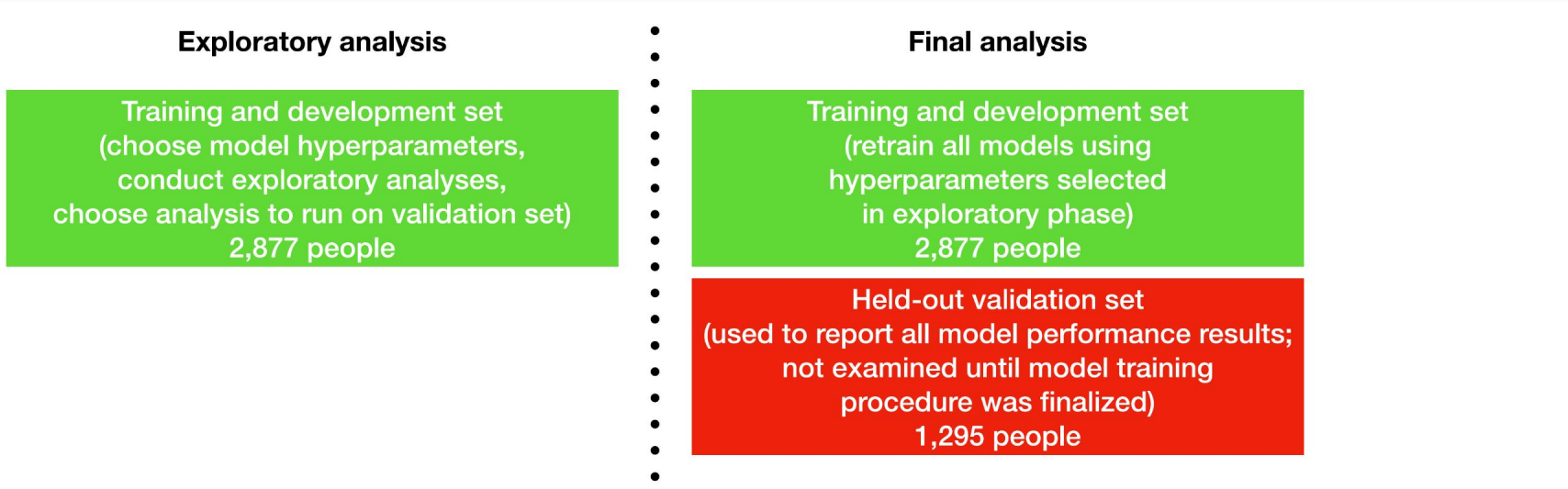
影像分级与疼痛差异

传统影像分级与患者疼痛存在失配。既往深度学习研究主要训练模型复现 KL 分级，把医师评分作为学习目标。

本文以患者自报疼痛训练影像模型，再比较两种严重程度指标对群体疼痛差异的解释能力，检验传统分级遗漏的信息。

方法

研究流程



Extended Data Fig. 2 | Analysis pipeline. Prior to conducting any analysis, 1,295 patients (red box) were reserved as a held-out validation set to assess final results. In the exploratory phase, the remaining patients were analyzed as follows: a training set was used to optimize model weights, and a development set to select model hyperparameters and conduct early stopping to avoid overfitting. The main analyses to run on the held-out validation set were determined prior to examining it, and the hyperparameters were finalized. In the final analysis, all models were retrained using the hyperparameters chosen in the exploratory phase, and model predictions were assessed on the 1,295 patients in the held-out validation set.

4172 名参与者，训练与开发 2877 人、最终验证 1295 人；按患者划分数据。

方法

以疼痛为学习目标

模型以膝关节 X 线片为输入，以患者的 KOOS 疼痛评分为主要学习目标。输出 ALG-P，概括影像中与疼痛相关的信息。

这一改动让患者体验参与定义疾病严重程度。随后把 ALG-P 与 KL 分级放进同一套差异分析，检验测量目标的改变带来了多少解释增量。

方法

影像输入与联合训练

每次输入双膝图像，统一目标膝位于图像右侧；使用 ResNet-18 和迁移学习，同时预测疼痛及 19 项影像特征。

疼痛预测误差决定模型选择，影像特征作为辅助学习任务。开发集用于调参，最终合并表现最好的 5 个模型。

方法

差异量化

先计算群体之间的平均疼痛差距，再分别加入 KLG 或 ALG-P 调整。调整后群体变量的回归系数，表示给定影像严重程度后的剩余差异。

解释比例 =（原始差距 − 调整后差距）÷ 原始差距。用同一问题和同一验证人群，比较两种测量方式保留了多少信息。

方法

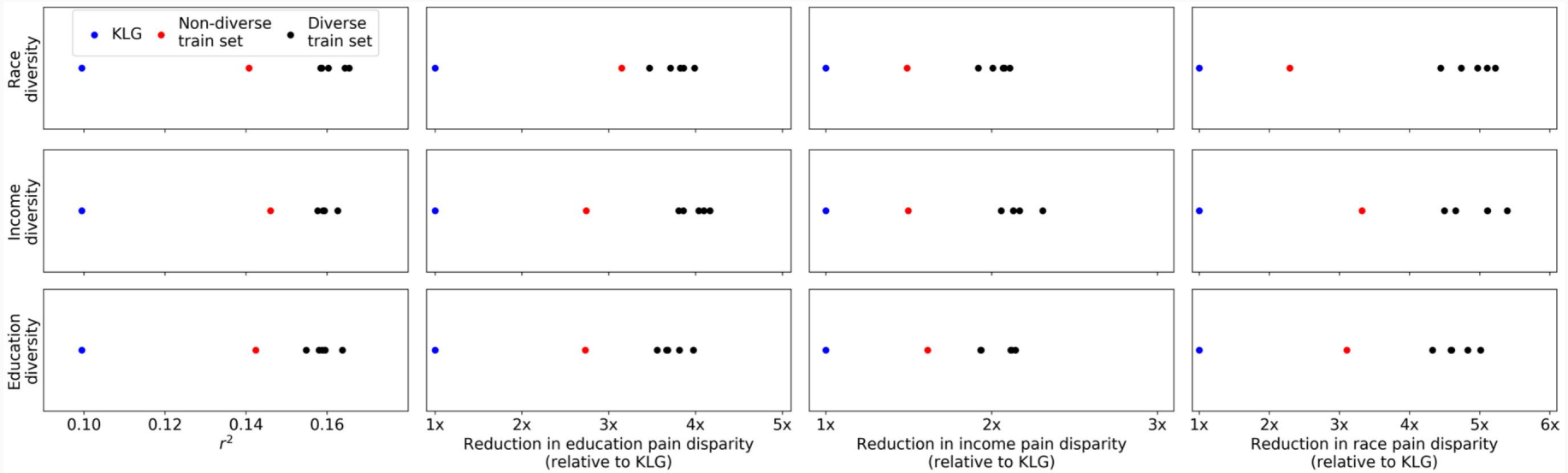
评分对齐

KLG 先在训练开发集中按五个等级分别拟合疼痛，再到验证集生成同尺度预测。
ALG-P 和 KLG 都接受独立验证集的检验。

进一步把 ALG-P 划成与 KLG 各等级人数一致的五档，比较相同粒度下的表现；并以交互项和分层加权方法检验线性模型的假设。

方法

等量数据消融



Extended Data Fig. 1 | The effect of dataset diversity on model performance. Each row of plots shows the effect of removing one minority group from the training set: from top, Black, lower-income, and lower-education patients. Each column of plots shows one metric: from left, R^2 in predicting KOOS pain score, and the reductions in the education, income, and racial pain disparities (relative to KLG). In each subplot, the blue dot shows, as a baseline, the performance of KLG. The red dot shows the performance of a neural network trained on a non-diverse training set, with all minority patients removed. The five black dots show the performance of neural networks trained on five diverse training sets of equal size, with five random subsets of non-minority patients removed; in all cases, the diverse training sets yield superior performance to non-diverse training sets of equal size.

固定训练集规模，比较移除某一人群的模型与保留多样性的模型；后者重复随机抽样 5 次，以分离样本数量与人群构成的影响。

方法

信号来源检验

逐区遮挡影像，观察疼痛预测变化；热区集中于目标膝及关节间隙周围。

再检验同一人群内部的预测、已知影像特征调整及留站点验证，逐层检查模型信息的来源。

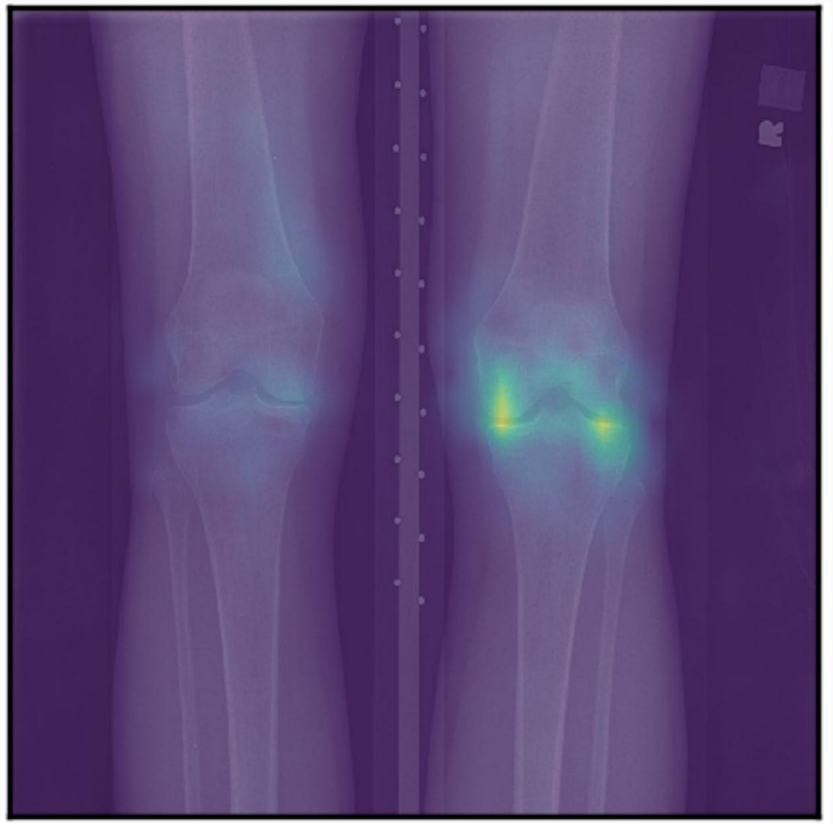


Fig. 1 | Heatmap of a representative X-ray image. The model's prediction target is the pain score in the knee appearing on the right side of the image. Regions that influence the prediction more strongly are shown in brighter colors.

结果

疼痛预测

Predictive performance for pain	KLG	ALG-P
Pearson R^2	0.10 (0.08, 0.13)	0.16 (0.13, 0.19)
Spearman R^2	0.08 (0.07, 0.11)	0.14 (0.11, 0.16)
RMSE	15.4 (14.7, 16.0)	14.9 (14.3, 15.4)
Mean absolute error	11.9 (11.5, 12.2)	11.3 (10.9, 11.7)
AUC (severe pain)	0.64 (0.62, 0.66)	0.69 (0.67, 0.71)

Extended Data Fig. 4 | Predictive performance for pain. 95% CIs are computed by cluster bootstrapping at the patient level.

验证集 Pearson R^2 为 0.16（ALG-P）与 0.10（KLG）；重度疼痛判别 AUC 为 0.69 与 0.64。模型捕捉了额外影像信号。

结果

差异解释

Table 2 | Reducing unexplained racial and socioeconomic disparities in pain

	Pain disparity (KOOS points) after controlling for			Reduction in pain disparity after controlling for		Ratio of reduction
	No severity measures	Radiographic severity (KLG)	Algorithmic severity (ALG-P)	Radiographic severity (KLG)	Algorithmic severity (ALG-P)	ALG-P to KLG
Race	10.6 (8.3, 12.9)	9.7 (7.4, 11.9)	6.1 (3.7, 8.3)	9% (3%, 16%)	43% (33%, 56%)	4.7 (3.2, 11.8)
Income	4.2 (2.8, 5.6)	3.5 (2.3, 4.9)	2.9 (1.6, 4.1)	16% (5%, 29%)	32% (18%, 50%)	2.0 (1.4, 4.4)
Education	5.3 (3.7, 6.7)	4.9 (3.5, 6.2)	3.7 (2.4, 5.0)	8% (−1%, 18%)	30% (18%, 44%)	3.6 (2.1, *)

The first three columns report racial and socioeconomic pain disparities (in KOOS points) without any controls for severity (first column, equivalent to the difference in mean pain scores between groups), when controlling for the clinician's severity measure KLG (second column) and when controlling for algorithmic severity measure ALG-P (third column). Controlling for either severity measure reduces racial and socioeconomic pain disparities, with the algorithmic severity measure achieving a larger reduction. The final three columns quantify the sizes of these reductions. The fourth column reports how much pain disparities are reduced by controlling for KLG, relative to controlling for no severity measures (that is, the reduction in the second column relative to the first). The fifth column reports how much pain disparities are reduced by controlling for ALG-P, relative to controlling for no severity measures (that is, the reduction in the third column relative to the first). The final column reports the ratio of the reductions in disparities by ALG-P versus KLG. In parentheses are 95% CIs computed by cluster bootstrapping at the patient level. The asterisk in the bottom-right entry (*) indicates that the upper limit of the CI was not defined, because the CI for the denominator included zero.

种族疼痛差距从 10.6 分，经 KLG 调整后为 9.7 分，经 ALG-P 调整后为 6.1 分；解释比例分别为 9% 与 43%。收入和教育分析方向一致。

结果

候选规则模拟

Table 3 | Potential eligibility for surgery: comparing KLG and ALG-P

	Knees potentially eligible for surgery (%)		Knees in severe pain and not eligible for surgery (%)	
	Using KLG	Using ALG-P	Using KLG	Using ALG-P
Black	11% (7%, 15%)	22% (17%, 27%)	51% (45%, 57%)	40% (34%, 46%)
Lower-income	10% (8%, 12%)	13% (10%, 15%)	36% (33%, 40%)	34% (31%, 38%)
Lower-education	9% (7%, 11%)	14% (11%, 16%)	38% (35%, 42%)	33% (30%, 37%)

在保持严重程度档位人数一致的筛选模拟中，黑人参与者的潜在置换候选膝比例由 11% 变为 22%；变化来自候选膝排序。

讨论

测量与临床判断

本文把“影像严重程度如何定义”纳入了研究问题。面向疼痛学习的指标改变了对群体差异的统计解释，提示传统分级遗漏了部分症状相关信息。

对骨科研究的启发，是让测量指标与患者关心的结局建立直接联系，并检验指标在人群之间的表现。症状、结构和治疗结局共同构成临床评价。

亮点

学习目标与验证思路

本文以患者报告的疼痛构建影像严重程度指标，把学习目标与患者体验直接联系起来。再将 ALG-P 与 KLG 放入同一差异分析，量化传统分级遗漏的信息。

验证设计围绕具体问题展开：等量数据消融分离样本数量与人群构成的影响，评分对齐控制尺度和粒度差别，分层与留站点分析检查模型在不同人群和采集环境中的表现。

多面板原图以固定布局和配色呈现控制条件、五次重复抽样与结果，使读者可以沿着研究问题理解证据，而每一步验证都有明确的解释对象。

局限

预测精度与应用距离

ALG-P 的 R^2 为 0.16，对个体疼痛的解释度较低。它增加了群体差异中的影像信息，但个体预测仍有较大误差，症状评估还包含影像之外的重要因素。

患者留出与留站点验证均来自 OAI 队列。队列的年龄、疾病风险与标准化拍片条件，界定了现有结果覆盖的人群和场景；新机构的病例结构与图像质量可能改变模型表现。

热图定位的是对模型预测敏感的区域，具体组织学对应尚待明确。置换分析以阈值规则下的候选膝比例为结局，实际决策、术后疼痛和并发症属于后续临床评价的核心内容。

感想

选题方向

可以选择结局导向的影像模型、训练样本代表性、我的课题中的测量偏差，把论文中的一个发现与自己的工作联系起来，再提出下一步要验证的问题。

填写时可以依次回应“最值得借鉴的方法”、“与我的课题的联系”、“我会怎样验证”，每个部分用一两句话展开。